VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"JnanaSangama", Belgaum- 590014, Karnataka.

LAB RECORD

on

Big Data Analytics (23CS6PCBDA)

Submitted by

Thilak K (1WA23CS021)

in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING

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B.M.S. College of Engineering

Bull Temple Road, Bangalore 560019
(Affiliated to Visvesvaraya Technological University, Belgaum)

Department of Computer Science and Engineering

CERTIFICATE

This is to certify that the Lab work entitled "Big Data Analytics" carried out by **Thilak K(1WA23CS021),** who is bonafide student of **B.M.S. College of Engineering.** It is in partial fulfilment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum during the year 2025. The Lab report has been approved as it satisfies the academic requirements in respect of a Big Data Analytics (23CS6PCBDA) work prescribed for the said degree.

Leelavathi. B
Assistant Professor
Department of CSE, BMSCE

Dr. Kavitha Sooda
Professor & HOD
Department of CSE, BMSCE

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		• Create a column family by name
		  o Employee-Info with attributes
		  o Emp_Id Primary Key, Emp_Name, Designation,
		  o Date_of_Joining, Salary, Dept_Name
		• Insert the values into the table in batch
		• Update Employee name and Department of EmpId 121
		• Sort the details of Employee records based on salary
		• Alter the schema of the table Employee_Info to add a column Projects which stores a set of Projects done by the corresponding Employee.
		• Update the altered table to add project names.
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3	13.02.26	Perform the following DB operations using Cassandra.
		• Create a keyspace by name Library
		• Create a column family by name Library-Info with attributes
		  o Stud_Id Primary Key,
		  o Counter_value of type Counter,
		  o Stud_Name, Book-Name, Book-Id,
		  o Date_of_issue
		• Insert the values into the table in batch
		• Display the details of the table created and increase the value of the counter
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Github Link: https://github.com/Thilak-K21/BDA-LAB

Course Outcomes (COs):

CO1	Apply the concept of NoSQL, Hadoop or Spark for a given task
CO2	Analyse big data analytics mechanisms that can be applied to obtain solution for a given problem.
CO3	Design and implement solutions using data analytics mechanisms for a given problem.

LABORATORY PROGRAM – 1

MongoDB- CRUD Operations Demonstration (Practice and Self Study)

COMMAND WITH OUTPUT - USING ATLAS

```
test> use BDA_LAB
switched to db BDA_LAB
BDA_LAB> db.createCollection("Employee")
{ ok: 1 }
BDA_LAB> db.Employee.insertOne({
...     Emp_Id: 101,
...     Name: "Manoj",
...     Dept: "CSE",
...     Salary: 50000
... })
{
  acknowledged: true,
  insertedId: ObjectId('69f5bb2311d23b4927149f48')
}
BDA_LAB> db.Employee.insertMany([
...     {Emp_Id: 102, Name: "Rahul", Dept: "ISE", Salary: 40000},
...     {Emp_Id: 103, Name: "Anjali", Dept: "ECE", Salary: 45000},
...     {Emp_Id: 104, Name: "Kiran", Dept: "CSE", Salary: 55000}
... ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69f5bb2a11d23b4927149f49'),
    '1': ObjectId('69f5bb2a11d23b4927149f4a'),
    '2': ObjectId('69f5bb2a11d23b4927149f4b')
  }
}
BDA_LAB> db.Employee.find()
[
  {
    _id: ObjectId('69f5bb2311d23b4927149f48'),
    Emp_Id: 101,
    Name: 'Manoj',
```

```
  }
}
BDA_LAB> db.Employee.find()
[
  {
    _id: ObjectId('69f5bb2311d23b4927149f48'),
    Emp_Id: 101,
    Name: 'Manoj',
    Dept: 'CSE',
    Salary: 50000
  },
  {
    _id: ObjectId('69f5bb2a11d23b4927149f49'),
    Emp_Id: 102,
    Name: 'Rahul',
    Dept: 'ISE',
    Salary: 40000
  },
  {
    _id: ObjectId('69f5bb2a11d23b4927149f4a'),
    Emp_Id: 103,
    Name: 'Anjali',
    Dept: 'ECE',
    Salary: 45000
  },
  {
    _id: ObjectId('69f5bb2a11d23b4927149f4b'),
    Emp_Id: 104,
    Name: 'Kiran',
    Dept: 'CSE',
    Salary: 55000
  }
]
BDA_LAB>
```

```
  },
  {
    _id: ObjectId('69f5bb2a11d23b4927149f4a'),
    Emp_Id: 103,
    Name: 'Anjali',
    Dept: 'ECE',
    Salary: 45000
  },
  {
    _id: ObjectId('69f5bb2a11d23b4927149f4b'),
    Emp_Id: 104,
    Name: 'Kiran',
    Dept: 'CSE',
    Salary: 55000
  }
]
BDA_LAB> db.Employee.find({Dept: "CSE"})
[
  {
    _id: ObjectId('69f5bb2311d23b4927149f48'),
    Emp_Id: 101,
    Name: 'Manoj',
    Dept: 'CSE',
    Salary: 50000
  },
  {
    _id: ObjectId('69f5bb2a11d23b4927149f4b'),
    Emp_Id: 104,
    Name: 'Kiran',
    Dept: 'CSE',
    Salary: 55000
  }
]
BDA_LAB>
```

```
BDA_LAB> db.Employee.updateOne(
...     {Emp_Id: 101},
...     {$set: {Salary: 60000}}
... )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
BDA_LAB> db.Employee.updateMany(
...     {Dept: "CSE"},
...     {$set: {Dept: "AI"}}
... )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
BDA_LAB>
```

```
BDA_LAB> db.Employee.deleteOne({Emp_Id: 102})
{ acknowledged: true, deletedCount: 1 }
BDA_LAB> db.Employee.deleteMany({Dept: "ECE"})
{ acknowledged: true, deletedCount: 1 }
BDA_LAB> show collections
Employee
BDA_LAB> db.Employee.drop()
true
BDA_LAB> db.dropDatabase()
{ ok: 1, dropped: 'BDA_LAB' }
BDA_LAB>
```

LABORATORY PROGRAM – 2

Perform the following DB operations using Cassandra

Questions:

a) Create a keyspace by name Employee
b) Create a column family by name

- Employee-Info with attributes
- Emp_Id Primary Key, Emp_Name,
- Designation, Date_of_Joining,
- Salary, Dept_Name

c) Insert the values into the table in batch
d) Update Employee name and Department of Emp-Id 121
e) Sort the details of Employee records based on salary
f) Alter the schema of the table Employee_Info to add a column Projects which stores a set of Projects done by the corresponding Employee.
g) Update the altered table to add project names.
h) Create a TTL of 15 seconds to display the values of Employees.

COMMAND WITH OUTPUT

```
cqlsh> CREATE KEYSPACE Employee
   ... WITH replication = {'class': 'SimpleStrategy', 'replication_factor': 1};

cqlsh>
cqlsh> USE Employee;
cqlsh:employee> CREATE TABLE Employee_Info (
            ...     Emp_Id INT PRIMARY KEY,
            ...     Emp_Name TEXT,
            ...     Designation TEXT,
            ...     Date_of_Joining DATE,
            ...     Salary INT,
            ...     Dept_Name TEXT
            ... );
cqlsh:employee> BEGIN BATCH
            ... INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name)
            ... VALUES (121, 'Manoj', 'Engineer', '2022-01-10', 50000, 'CSE');
            ...
            ... INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name)
            ... VALUES (122, 'Rahul', 'Manager', '2021-05-15', 70000, 'ISE');
            ...
            ... INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name)
            ... VALUES (123, 'Anjali', 'Analyst', '2023-03-20', 45000, 'ECE');
            ... APPLY BATCH;
cqlsh:employee> UPDATE Employee_Info
            ... SET Emp_Name = 'Manoj Kumar', Dept_Name = 'AI'
            ... WHERE Emp_Id = 121;
cqlsh:employee> SELECT * FROM Employee_Info ORDER BY Salary DESC ALLOW FILTERING;
InvalidRequest: Error from server: code=2200 [Invalid query] message="ORDER BY is only supported when the partition key is restricted b
y an EQ or an IN."
cqlsh:employee> ALTER TABLE Employee_Info
            ... ADD Projects SET<TEXT>;
cqlsh:employee> UPDATE Employee_Info
            ... SET Projects = {'Hadoop', 'Spark'}
            ... WHERE Emp_Id = 121;
cqlsh:employee>
cqlsh:employee> UPDATE Employee_Info
            ... SET Projects = {'ML', 'Data Mining'}
            ... WHERE Emp_Id = 122;
cqlsh:employee>
```

```
cqlsh:employee> INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name)
             ... VALUES (130, 'TempUser', 'Intern', '2024-01-01', 20000, 'CSE')
             ... USING TTL 15;
cqlsh:employee> SELECT * FROM Employee_Info;

 emp_id | date_of_joining | dept_name | designation | emp_name    | projects                | salary
--------+-----------------+-----------+-------------+-------------+-------------------------+--------
    123 |      2023-03-20 |       ECE |     Analyst |      Anjali |                    null |  45000
    130 |      2024-01-01 |       CSE |      Intern |    TempUser |                    null |  20000
    122 |      2021-05-15 |       ISE |     Manager |       Rahul | {'Data Mining', 'ML'} |  70000
    121 |      2022-01-10 |        AI |    Engineer | Manoj Kumar |   {'Hadoop', 'Spark'} |  50000

(4 rows)
cqlsh:employee> SELECT * FROM Employee_Info;

 emp_id | date_of_joining | dept_name | designation | emp_name    | projects                | salary
--------+-----------------+-----------+-------------+-------------+-------------------------+--------
    123 |      2023-03-20 |       ECE |     Analyst |      Anjali |                    null |  45000
    130 |      2024-01-01 |       CSE |      Intern |    TempUser |                    null |  20000
    122 |      2021-05-15 |       ISE |     Manager |       Rahul | {'Data Mining', 'ML'} |  70000
    121 |      2022-01-10 |        AI |    Engineer | Manoj Kumar |   {'Hadoop', 'Spark'} |  50000

(4 rows)
cqlsh:employee>
```

LABORATORY PROGRAM – 3

Perform the following DB operations using Cassandra

Questions:

a) Create a keyspace by name Library
b) Create a column family by name Library-Info with attributes

- Stud_Id Primary Key,
- Counter_value of type Counter,
- Stud_Name, Book-Name, Book-Id,
- Date_of_issue

c) Insert the values into the table in batch
d) Display the details of the table created and increase the value of the counter
e) Write a query to show that a student with id 112 has taken a book "BDA" 2 times.
f) Export the created column to a csv file
g) Import a given csv dataset from local file system into Cassandra column family

COMMAND WITH OUTPUT

```
cqlsh> CREATE KEYSPACE Library
   ... WITH replication = {'class': 'SimpleStrategy', 'replication_factor': 1};
cqlsh> USE Library;
cqlsh:library> CREATE TABLE Library_Info (
           ...     Stud_Id INT PRIMARY KEY,
           ...     Stud_Name TEXT,
           ...     Book_Name TEXT,
           ...     Book_Id INT,
           ...     Date_of_issue DATE
           ... );
cqlsh:library> CREATE TABLE Book_Counter (
           ...     Stud_Id INT,
           ...     Book_Name TEXT,
           ...     Counter_Value COUNTER,
           ...     PRIMARY KEY (Stud_Id, Book_Name)
           ... );
cqlsh:library> BEGIN BATCH
           ... INSERT INTO Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id, Date_of_issue)
           ... VALUES (111, 'Manoj', 'BDA', 201, '2024-04-01');
           ...
           ... INSERT INTO Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id, Date_of_issue)
           ... VALUES (112, 'Rahul', 'BDA', 202, '2024-04-02');
           ...
           ... INSERT INTO Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id, Date_of_issue)
           ... VALUES (113, 'Anjali', 'DBMS', 203, '2024-04-03');
           ... APPLY BATCH;
cqlsh:library> UPDATE Book_Counter
           ... SET Counter_Value = Counter_Value + 1
           ... WHERE Stud_Id = 112 AND Book_Name = 'BDA';
cqlsh:library>
cqlsh:library> UPDATE Book_Counter
           ... SET Counter_Value = Counter_Value + 1
           ... WHERE Stud_Id = 112 AND Book_Name = 'BDA';
cqlsh:library>
```

```
cqlsh:library> SELECT * FROM Library_Info;

 stud_id | book_id | book_name | date_of_issue | stud_name
---------+---------+-----------+---------------+----------
     111 |     201 |       BDA |    2024-04-01 |     Manoj
     113 |     203 |      DBMS |    2024-04-03 |    Anjali
     112 |     202 |       BDA |    2024-04-02 |     Rahul

(3 rows)
cqlsh:library> SELECT * FROM Book_Counter;

 stud_id | book_name | counter_value
---------+-----------+---------------
     112 |       BDA |             2

(1 rows)
cqlsh:library> SELECT * FROM Book_Counter
           ... WHERE Stud_Id = 112 AND Book_Name = 'BDA';

 stud_id | book_name | counter_value
---------+-----------+---------------
     112 |       BDA |             2

(1 rows)
cqlsh:library> cqlsh -e "COPY Library.Library_Info TO 'library.csv' WITH HEADER = TRUE;"
           ...
cqlsh:library>
```

```
┌──(thilak㉿kali)-[~]
└─$ cqlsh -e "COPY Library.Library_Info TO 'library.csv' WITH HEADER = TRUE;"
/usr/bin/cqlsh.py:473: DeprecationWarning: Legacy execution parameters will be removed in 4.0. Consider using execution profil
es.
/usr/bin/cqlsh.py:503: DeprecationWarning: Setting the consistency level at the session level will be removed in 4.0. Consider
 using execution profiles and setting the desired consistency level to the EXEC_PROFILE_DEFAULT profile.
Using 5 child processes
/usr/bin/cqlsh.py:1555: DeprecationWarning: setDaemon() is deprecated, set the daemon attribute instead

Starting copy of library.library_info with columns [stud_id, book_id, book_name, date_of_issue, stud_name].
Processed: 3 rows; Rate:      12 rows/s; Avg. rate:      12 rows/s
3 rows exported to 1 files in 0.279 seconds.

┌──(thilak㉿kali)-[~]
└─$ cqlsh -e "COPY Library.Library_Info FROM 'library.csv' WITH HEADER = TRUE;"
/usr/bin/cqlsh.py:473: DeprecationWarning: Legacy execution parameters will be removed in 4.0. Consider using execution profil
es.
/usr/bin/cqlsh.py:503: DeprecationWarning: Setting the consistency level at the session level will be removed in 4.0. Consider
 using execution profiles and setting the desired consistency level to the EXEC_PROFILE_DEFAULT profile.
Using 5 child processes
/usr/bin/cqlsh.py:1553: DeprecationWarning: setDaemon() is deprecated, set the daemon attribute instead

Starting copy of library.library_info with columns [stud_id, book_id, book_name, date_of_issue, stud_name].
Processed: 3 rows; Rate:       4 rows/s; Avg. rate:       6 rows/s
3 rows imported from 1 files in 0.482 seconds (0 skipped).

┌──(thilak㉿kali)-[~]
└─$
```

LABORATORY PROGRAM – 4

Execution of HDFS Commands for interaction with Hadoop Environment. (Minimum 10 commands to be executed)

COMMAND WITH OUTPUT

```
┌──(thilak㉿kali)-[~/hadoop/sbin]
└─$ hdfs dfs -mkdir /bda

┌──(thilak㉿kali)-[~/hadoop/sbin]
└─$ hdfs dfs -mkdir /bda/input

┌──(thilak㉿kali)-[~/hadoop/sbin]
└─$ hdfs dfs -ls /
Found 1 items
drwxr-xr-x   - thilak supergroup          0 2026-05-02 14:45 /bda

┌──(thilak㉿kali)-[~/hadoop/sbin]
└─$ nano sample.txt

┌──(thilak㉿kali)-[~/hadoop/sbin]
└─$ hdfs dfs -put sample.txt /bda/input/

┌──(thilak㉿kali)-[~/hadoop/sbin]
└─$ hdfs dfs -cat /bda/input/sample.txt
Hello Hadoop
Big Data Analytics

┌──(thilak㉿kali)-[~/hadoop/sbin]
└─$ hdfs dfs -cp /bda/input/sample.txt /bda/sample_copy.txt

┌──(thilak㉿kali)-[~/hadoop/sbin]
└─$ hdfs dfs -mv /bda/sample_copy.txt /bda/input/

┌──(thilak㉿kali)-[~/hadoop/sbin]
└─$ hdfs dfs -get /bda/input/sample.txt
get: `sample.txt': File exists

┌──(thilak㉿kali)-[~/hadoop/sbin]
└─$ hdfs dfs -rm /bda/input/sample.txt
Deleted /bda/input/sample.txt

┌──(thilak㉿kali)-[~/hadoop/sbin]
└─$ hdfs dfs -rm /bda/input/sample.txt
rm: `/bda/input/sample.txt': No such file or directory

┌──(thilak㉿kali)-[~/hadoop/sbin]
└─$ hdfs dfs -rm -r /bda
Deleted /bda
```


```
┌──(thilak㉿kali)-[~/hadoop/sbin]
└─$ hdfs dfs -mkdir /bda

┌──(thilak㉿kali)-[~/hadoop/sbin]
└─$ hdfs dfs -mkdir /bda/input

┌──(thilak㉿kali)-[~/hadoop/sbin]
└─$ hdfs dfs -ls /
Found 1 items
drwxr-xr-x   - thilak supergroup          0 2026-05-02 14:45 /bda

┌──(thilak㉿kali)-[~/hadoop/sbin]
└─$ nano sample.txt

┌──(thilak㉿kali)-[~/hadoop/sbin]
└─$ hdfs dfs -put sample.txt /bda/input/

┌──(thilak㉿kali)-[~/hadoop/sbin]
└─$ hdfs dfs -cat /bda/input/sample.txt
Hello Hadoop
Big Data Analytics

┌──(thilak㉿kali)-[~/hadoop/sbin]
└─$ hdfs dfs -cp /bda/input/sample.txt /bda/sample_copy.txt

┌──(thilak㉿kali)-[~/hadoop/sbin]
└─$ hdfs dfs -mv /bda/sample_copy.txt /bda/input/

┌──(thilak㉿kali)-[~/hadoop/sbin]
└─$ hdfs dfs -get /bda/input/sample.txt
get: `sample.txt': File exists

┌──(thilak㉿kali)-[~/hadoop/sbin]
└─$ hdfs dfs -rm /bda/input/sample.txt
Deleted /bda/input/sample.txt

┌──(thilak㉿kali)-[~/hadoop/sbin]
└─$ hdfs dfs -rm /bda/input/sample.txt
rm: `/bda/input/sample.txt': No such file or directory

┌──(thilak㉿kali)-[~/hadoop/sbin]
└─$ hdfs dfs -rm -r /bda
Deleted /bda
```

LABORATORY PROGRAM – 5
Implement Wordcount program on Hadoop framework

```
import java.io.Exception
import org.apache.hadoop.conf.Configured;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.FileInputFormat;
import org.apache.hadoop.mapred.FileOutputFormat;
import org.apache.hadoop.mapred.JobClient;
import org.apache.hadoop.mapred.JobConf;
import org.apache.hadoop.util.Tool;
import org.apache.hadoop.util.ToolRunner;

public class WCDriver extends Configured implements Tool {

    public int run(String[] args) throws IOException {

        if (args.length < 2) {
            System.out.println("Please give valid inputs");
            return -1;
        }

        JobConf conf = new JobConf(WCDriver.class);
        conf.setJobName("WordCount");

        FileInputFormat.setInputPaths(conf, new Path(args[0]));
        FileOutputFormat.setOutputPath(conf, new Path(args[1]));

        conf.setMapperClass(WCMapper.class);
        conf.setReducerClass(WCReducer.class);

        conf.setMapOutputKeyClass(Text.class);
        conf.setMapOutputValueClass(IntWritable.class);

        conf.setOutputKeyClass(Text.class);
        conf.setOutputValueClass(IntWritable.class);

        JobClient.runJob(conf);
        return 0;
    }

    // Main Method
```

```
    public static void main(String[] args) throws Exception {
        int exitCode = ToolRunner.run(new WCDriver(), args);
        System.out.println("Job Exit Code: " + exitCode);
    }
  }}
```

Mapper Code:

```
// Importing libraries
import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.MapReduceBase;
import org.apache.hadoop.mapred.Mapper;
import org.apache.hadoop.mapred.OutputCollector;
import org.apache.hadoop.mapred.Reporter;

public class WCMapper extends MapReduceBase
    implements Mapper<LongWritable, Text, Text, IntWritable> {

  // Map function
  public void map(LongWritable key, Text value,
            OutputCollector<Text, IntWritable> output,
            Reporter reporter) throws IOException {

    String line = value.toString();

    // Splitting the line on whitespace
    for (String word : line.split("\\s+")) {
      if (word.length() > 0) {
        output.collect(new Text(word), new IntWritable(1));
      }
    }
  }
}
```

Reducer Code

```
// Importing libraries
import java.io.IOException;
import java.util.Iterator;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.MapReduceBase;
import org.apache.hadoop.mapred.OutputCollector;
import org.apache.hadoop.mapred.Reducer;
import org.apache.hadoop.mapred.Reporter;
```

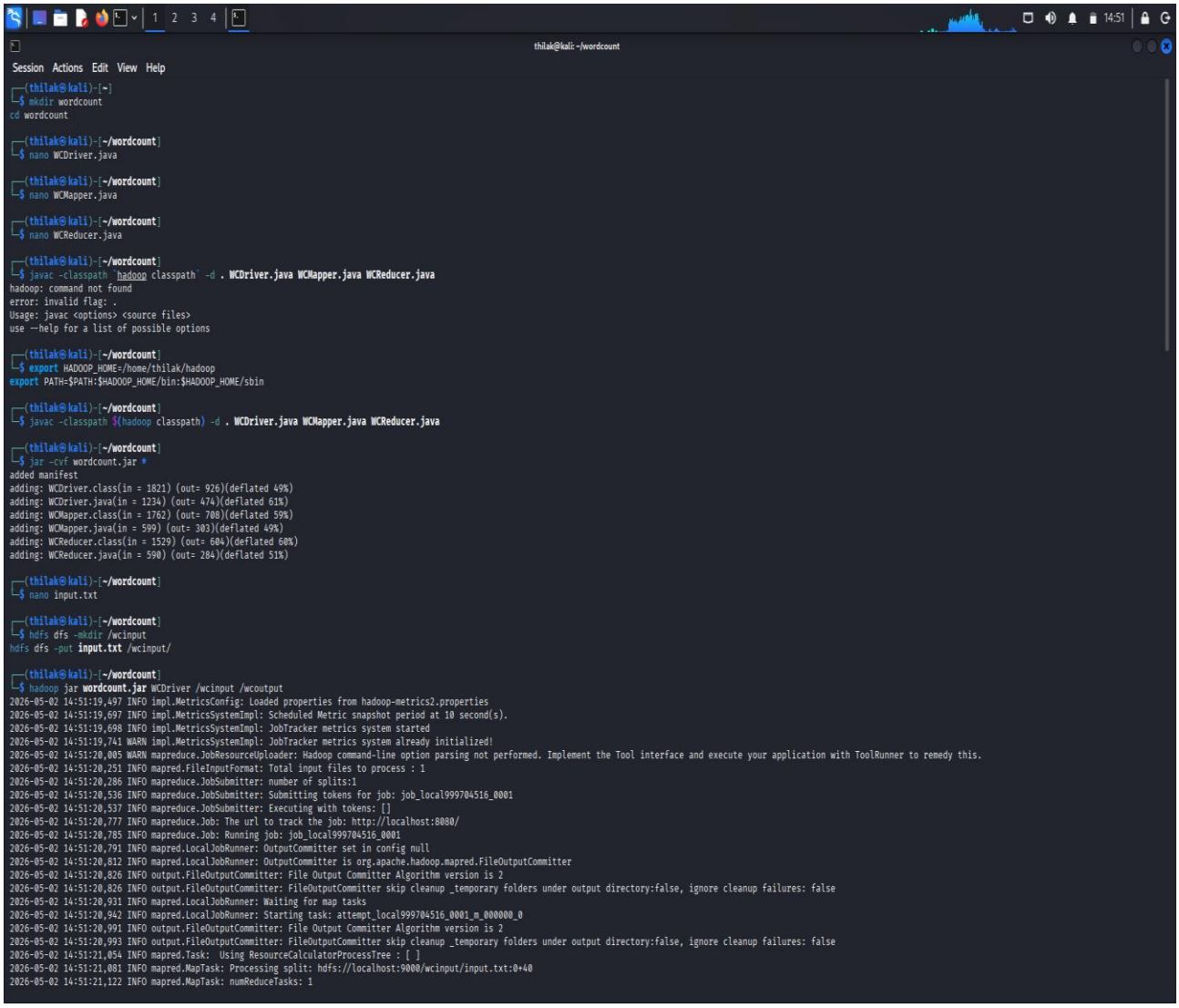
```
public class WCReducer extends MapReduceBase
      implements Reducer<Text, IntWritable, Text, IntWritable> {

   // Reduce function
   public void reduce(Text key, Iterator<IntWritable> values,
                      OutputCollector<Text, IntWritable> output,
                      Reporter reporter) throws IOException {

      int count = 0;

      // Counting the frequency of each word
      while (values.hasNext()) {
         count += values.next().get();
      }

      output.collect(key, new IntWritable(count));
   }
}
```



```
Session  Actions  Edit  View  Help
                HDFS: Number of bytes read=80
                HDFS: Number of bytes written=30
                HDFS: Number of read operations=15
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=4
                HDFS: Number of bytes read erasure-coded=0
        Map-Reduce Framework
                Map input records=3
                Map output records=7
                Map output bytes=68
                Map output materialized bytes=88
                Input split bytes=91
                Combine input records=0
                Combine output records=0
                Reduce input groups=4
                Reduce shuffle bytes=88
                Reduce input records=7
                Reduce output records=4
                Spilled Records=14
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=12
                Total committed heap usage (bytes)=432013312
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=40
        File Output Format Counters
                Bytes Written=30

┌──(thilak㉿kali)-[~/wordcount]
└─$ hdfs dfs -cat /wcoutput/part-00000
big     1
data    2
hadoop  2
hello   2

┌──(thilak㉿kali)-[~/wordcount]
└─$
```

LABORATORY PROGRAM – 6

Implement Weather program on Hadoop framework

Questions:

From the following link extract the weather data
https://github.com/tomwhite/hadoopbook/tree/master/input/ncdc/all
a) Create a MapReduce program to find average temperature for each year from NCDC data set.
b) find the mean max temperature for every month.

CODE, COMMAND WITH OUTPUT – A
Driver Code

```
package temp;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class AverageDriver {

   public static void main(String[] args) throws Exception {

      if (args.length != 2) {
         System.err.println("Please enter both input and output parameters.");
         System.exit(-1);
      }

      // Creating a configuration and job instance
      Configuration conf = new Configuration();
      Job job = Job.getInstance(conf, "Average Calculation");
      job.setJarByClass(AverageDriver.class);

      // Input and output paths
      FileInputFormat.addInputPath(job, new Path(args[0]));
      FileOutputFormat.setOutputPath(job, new Path(args[1]));

      // Setting mapper and reducer classes
      job.setMapperClass(AverageMapper.class);
      job.setReducerClass(AverageReducer.class);
```

```
        // Output key and value types
        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(IntWritable.class);

        // Submitting the job and waiting for it to complete
        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}
```

MAPPER CODE

```
package temp;


import java.io.IOException;


import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;


public class AverageMapper extends Mapper<LongWritable, Text, Text, IntWritable> {


    public static final int MISSING = 9999;


    @Override
    public void map(LongWritable key, Text value, Context context)
            throws IOException, InterruptedException {


        String line = value.toString();


        // Extract year from fixed position
        String year = line.substring(15, 19);
        int temperature;


        // Determine if there's a '+' sign
        if (line.charAt(87) == '+') {
```

```
            temperature = Integer.parseInt(line.substring(88, 92));
        } else {
            temperature = Integer.parseInt(line.substring(87, 92));
        }

        // Quality check character
        String quality = line.substring(92, 93);

        // Only emit if data is valid
        if (temperature != MISSING && quality.matches("[01459]")) {
            context.write(new Text(year), new IntWritable(temperature));
        }
    }
```

REDUCER CODE

```
package temp;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class AverageReducer extends Reducer<Text, IntWritable, Text, IntWritable> {

    @Override
    public void reduce(Text key, Iterable<IntWritable> values, Context context)
            throws IOException, InterruptedException {

        int sumTemp = 0;
        int count = 0;
```

```
        for (IntWritable value : values) {
            sumTemp += value.get();
            count++;
        }

        if (count > 0) {
            int average = sumTemp / count;
            context.write(key, new IntWritable(average));
        }
    }
```

```
thilak@kali: ~/weather
Session  Actions  Edit  View  Help
┌──(thilak㉿kali)-[~]
└─$ mkdir weather
cd weather

┌──(thilak㉿kali)-[~/weather]
└─$ nano AverageDriver.java

┌──(thilak㉿kali)-[~/weather]
└─$ nano AverageMapper.java

┌──(thilak㉿kali)-[~/weather]
└─$ nano AverageReducer.java

┌──(thilak㉿kali)-[~/weather]
└─$ javac -classpath $(hadoop classpath) -d . *.java
hadoop: command not found
error: invalid flag: .
Usage: javac <options> <source files>
use --help for a list of possible options

┌──(thilak㉿kali)-[~/weather]
└─$ export HADOOP_HOME=/home/thilak/hadoop
export PATH=$PATH:$HADOOP_HOME/bin:$HADOOP_HOME/sbin

┌──(thilak㉿kali)-[~/weather]
└─$ javac -classpath $(hadoop classpath) -d . *.java

┌──(thilak㉿kali)-[~/weather]
└─$ jar -cvf weather.jar *
added manifest
adding: AverageDriver.class(in = 1369) (out= 748)(deflated 45%)
adding: AverageDriver.java(in = 919) (out= 374)(deflated 59%)
adding: AverageMapper.class(in = 1708) (out= 697)(deflated 59%)
adding: AverageMapper.java(in = 657) (out= 315)(deflated 52%)
adding: AverageReducer.class(in = 1609) (out= 677)(deflated 57%)
adding: AverageReducer.java(in = 527) (out= 267)(deflated 49%)

┌──(thilak㉿kali)-[~/weather]
└─$ nano weather.txt

┌──(thilak㉿kali)-[~/weather]
└─$ hdfs dfs -mkdir /weatherinput
hdfs dfs -put weather.txt /weatherinput/

┌──(thilak㉿kali)-[~/weather]
└─$ hadoop jar weather.jar AverageDriver /weatherinput /weatheroutput
2026-05-02 14:55:24,936 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2026-05-02 14:55:25,104 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2026-05-02 14:55:25,104 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2026-05-02 14:55:25,313 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-05-02 14:55:25,575 INFO input.FileInputFormat: Total input files to process : 1
2026-05-02 14:55:25,609 INFO mapreduce.JobSubmitter: number of splits:1
2026-05-02 14:55:25,836 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local1283312055_0001
2026-05-02 14:55:25,836 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-05-02 14:55:26,059 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2026-05-02 14:55:26,062 INFO mapreduce.Job: Running job: job_local1283312055_0001
2026-05-02 14:55:26,071 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2026-05-02 14:55:26,109 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-05-02 14:55:26,112 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-05-02 14:55:26,112 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2026-05-02 14:55:26,117 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter
2026-05-02 14:55:26,222 INFO mapred.LocalJobRunner: Waiting for map tasks
2026-05-02 14:55:26,227 INFO mapred.LocalJobRunner: Starting task: attempt_local1283312055_0001_m_000000_0
2026-05-02 14:55:26,280 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-05-02 14:55:26,282 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-05-02 14:55:26,283 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2026-05-02 14:55:26,341 INFO mapred.Task:  Using ResourceCalculatorProcessTree : [ ]
2026-05-02 14:55:26,351 INFO mapred.MapTask: Processing split: hdfs://localhost:9000/weatherinput/weather.txt:0+282
```

```
thilak@kali: ~/weather
Session  Actions  Edit  View  Help
                FILE: Number of write operations=0
                HDFS: Number of bytes read=564
                HDFS: Number of bytes written=22
                HDFS: Number of read operations=15
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=4
                HDFS: Number of bytes read erasure-coded=0
        Map-Reduce Framework
                Map input records=3
                Map output records=3
                Map output bytes=27
                Map output materialized bytes=39
                Input split bytes=111
                Combine input records=0
                Combine output records=0
                Reduce input groups=2
                Reduce shuffle bytes=39
                Reduce input records=3
                Reduce output records=2
                Spilled Records=6
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=15
                Total committed heap usage (bytes)=419430400
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=282
        File Output Format Counters
                Bytes Written=22

┌──(thilak㉿kali)-[~/weather]
└─$ hdfs dfs -cat /weatheroutput/part-00000
cat: `/weatheroutput/part-00000': No such file or directory

┌──(thilak㉿kali)-[~/weather]
└─$ hdfs dfs -ls /weatheroutput
Found 2 items
-rw-r--r--   3 thilak supergroup          0 2026-05-02 14:57 /weatheroutput/_SUCCESS
-rw-r--r--   3 thilak supergroup         22 2026-05-02 14:57 /weatheroutput/part-r-00000

┌──(thilak㉿kali)-[~/weather]
└─$ hdfs dfs -cat /weatheroutput/part-r-00000
5202    89012
5678    89012

┌──(thilak㉿kali)-[~/weather]
└─$
```

LABORATORY PROGRAM – 7

For a given Text file, Create a Map Reduce program to sort the content in an alphabetic order listing only top 10 maximum occurrences of words.

```
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class TopNDriver {

    public static void main(String[] args) throws Exception {

        if (args.length != 3) {
            System.err.println("Usage: TopNDriver <in> <temp-out> <final-out>");
            System.exit(2);
        }

        Configuration conf = new Configuration();

        // === Job 1: Word Count ===
        Job wcJob = Job.getInstance(conf, "word count");
        wcJob.setJarByClass(TopNDriver.class);
        wcJob.setMapperClass(WordCountMapper.class);
        wcJob.setCombinerClass(WordCountReducer.class);
        wcJob.setReducerClass(WordCountReducer.class);
        wcJob.setOutputKeyClass(Text.class);
        wcJob.setOutputValueClass(IntWritable.class);

        FileInputFormat.addInputPath(wcJob, new Path(args[0]));
        Path tempDir = new Path(args[1]);
        FileOutputFormat.setOutputPath(wcJob, tempDir);

        if (!wcJob.waitForCompletion(true)) {
            System.exit(1);
        }

        // === Job 2: Top N ===
```

```
        Job topJob = Job.getInstance(conf, "top 10 words");
        topJob.setJarByClass(TopNDriver.class);
        topJob.setMapperClass(TopNMapper.class);
        topJob.setReducerClass(TopNReducer.class);
        topJob.setMapOutputKeyClass(IntWritable.class);
        topJob.setMapOutputValueClass(Text.class);
        topJob.setOutputKeyClass(Text.class);
        topJob.setOutputValueClass(IntWritable.class);

        FileInputFormat.addInputPath(topJob, tempDir);
        FileOutputFormat.setOutputPath(topJob, new Path(args[2]));

        System.exit(topJob.waitForCompletion(true) ? 0 : 1);
    }
}
```

Mapper Code (WordCountMapper.java)

```
package samples.topn;

import java.io.IOException;
import java.util.StringTokenizer;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class WordCountMapper extends Mapper<Object, Text, Text, IntWritable> {

    private static final IntWritable ONE = new IntWritable(1);
    private Text word = new Text();

    // characters to normalize into spaces
    private String tokens = "[_|$#<>\\^=\\[\\]\\*/\\\\,;,.\\-:()?!\"']";

    @Override
    protected void map(Object key, Text value, Context context)
            throws IOException, InterruptedException {

        // clean & tokenize
        String clean = value.toString()
                .toLowerCase()
                .replaceAll(tokens, " ");

        StringTokenizer itr = new StringTokenizer(clean);
```

```
        while (itr.hasMoreTokens()) {
            word.set(itr.nextToken().trim());
            context.write(word, ONE);
        }
    }
}
```

Mapper Code (TopNMapper.java)

```
package samples.topn;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class TopNMapper extends Mapper<Object, Text, IntWritable, Text> {

    private IntWritable count = new IntWritable();
    private Text word = new Text();

    @Override
    protected void map(Object key, Text value, Context context)
            throws IOException, InterruptedException {

        // input line: word \t count
        String[] parts = value.toString().split("\\t");

        if (parts.length == 2) {
            word.set(parts[0]);
            count.set(Integer.parseInt(parts[1]));

            // emit count → word, so Hadoop sorts by count
            context.write(count, word);
        }
    }
}
```

Reducer Code (WordCountReducer.java)

```
package samples.topn;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;
```

```
public class WordCountReducer extends Reducer<Text, IntWritable, Text, IntWritable> {

    @Override
    protected void reduce(Text key, Iterable<IntWritable> values, Context context)
            throws IOException, InterruptedException {

        int sum = 0;

        for (IntWritable val : values) {
            sum += val.get();
        }

        context.write(key, new IntWritable(sum));
    }
}
```

Reducer Code (TopNReducer.java)

```
package samples.topn;

import java.io.IOException;
import java.util.ArrayList;
import java.util.Collections;
import java.util.List;
import java.util.Map;
import java.util.TreeMap;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class TopNReducer extends Reducer<IntWritable, Text, Text, IntWritable> {

    // TreeMap with descending order of keys (counts)
    private TreeMap<Integer, List<String>> countMap =
            new TreeMap<>(Collections.reverseOrder());

    @Override
    protected void reduce(IntWritable key, Iterable<Text> values, Context context)
            throws IOException, InterruptedException {

        int cnt = key.get();

        List<String> words = countMap.getOrDefault(cnt, new ArrayList<>());
```

```
        for (Text w : values) {
            words.add(w.toString());
        }

        countMap.put(cnt, words);
    }

    @Override
    protected void cleanup(Context context)
            throws IOException, InterruptedException {

        // collect top 10 word→count pairs
        List<WordCount> topList = new ArrayList<>();
        int seen = 0;

        for (Map.Entry<Integer, List<String>> entry : countMap.entrySet()) {
            int cnt = entry.getKey();

            for (String w : entry.getValue()) {
                topList.add(new WordCount(w, cnt));
                seen++;

                if (seen == 10) break;
            }

            if (seen == 10) break;
        }

        // emit results
        for (WordCount wc : topList) {
            context.write(new Text(wc.getWord()), new IntWritable(wc.getCount()));
        }
    }
}
        // sort these 10 entries alphabetically by word
        Collections.sort(topList, (a, b) -> a.word.compareTo(b.word));

        // emit final top 10 in alphabetical order
        for (WordCount wc : topList) {
            context.write(new Text(wc.word), new IntWritable(wc.count));
        }
    }

    // helper class
```

```
    private static class WordCount {
        String word;
        int count;

        WordCount(String w, int c) {
            word = w;
            count = c;
        }
    }
}
```

```
┌──(thilak㉿kali)-[~]
└─$ mkdir top10
cd top10

┌──(thilak㉿kali)-[~/top10]
└─$ nano TopDriver.java

┌──(thilak㉿kali)-[~/top10]
└─$ nano TopMapper.java

┌──(thilak㉿kali)-[~/top10]
└─$ nano TopReducer.java

┌──(thilak㉿kali)-[~/top10]
└─$ javac -classpath $(hadoop classpath) -d . *.java
hadoop: command not found
error: invalid flag: .
Usage: javac <options> <source files>
use --help for a list of possible options

┌──(thilak㉿kali)-[~/top10]
└─$ export HADOOP_HOME=/home/thilak/hadoop
export PATH=$PATH:$HADOOP_HOME/bin:$HADOOP_HOME/sbin

┌──(thilak㉿kali)-[~/top10]
└─$ javac -classpath $(hadoop classpath) -d . *.java

┌──(thilak㉿kali)-[~/top10]
└─$ jar -cvf top10.jar *
added manifest
adding: TopDriver.class(in = 1343) (out= 743)(deflated 44%)
adding: TopDriver.java(in = 893) (out= 367)(deflated 58%)
adding: TopMapper.class(in = 1709) (out= 714)(deflated 58%)
adding: TopMapper.java(in = 551) (out= 291)(deflated 47%)
adding: TopReducer.class(in = 3664) (out= 1513)(deflated 58%)
adding: TopReducer.java(in = 1199) (out= 504)(deflated 57%)

┌──(thilak㉿kali)-[~/top10]
└─$
```

```
Session  Actions  Edit  View  Help
                HDFS: Number of bytes written=44
                HDFS: Number of read operations=15
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=4
                HDFS: Number of bytes read erasure-coded=0
        Map-Reduce Framework
                Map input records=4
                Map output records=13
                Map output bytes=125
                Map output materialized bytes=157
                Input split bytes=105
                Combine input records=0
                Combine output records=0
                Reduce input groups=5
                Reduce shuffle bytes=157
                Reduce input records=13
                Reduce output records=5
                Spilled Records=26
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=15
                Total committed heap usage (bytes)=434110464
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=73
        File Output Format Counters
                Bytes Written=44

┌──(thilak㉿kali)-[~/top10]
└─$ hdfs dfs -cat /topoutput/part-r-00000
analytics       1
big     4
data    5
hadoop  2
science 1

┌──(thilak㉿kali)-[~/top10]
└─$
```

LABORATORY PROGRAM – 8

Write a Scala program to print numbers from 1 to 100 using for loop.

CODE, COMMAND WITH OUTPUT

```
┌──(thilak㉿kali)-[~]
└─$ scala -version
Scala code runner version 2.11.12 -- Copyright 2002-2017, LAMP/EPFL

┌──(thilak㉿kali)-[~]
└─$ nano numbers.scala

┌──(thilak㉿kali)-[~]
└─$ scalac numbers.scala

┌──(thilak㉿kali)-[~]
└─$ scala Numbers
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
```

LABORATORY PROGRAM – 9

Using RDD and FlatMap count how many times each word appears in a file and write out a list of words whose count is strictly greater than 4 using Spark.

CODE, COMMAND WITH OUTPUT

```
>>> rdd = spark.sparkContext.textFile("input.txt")
>>> words = rdd.flatMap(lambda line: line.split(" "))
>>> pairs = words.map(lambda word: (word, 1))
>>> counts = pairs.reduceByKey(lambda a, b: a + b)
>>> result = counts.filter(lambda x: x[1] > 4)
>>> result.collect()
[('hello', 5), ('spark', 6)]
>>>
```

LABORATORY PROGRAM – 10

Write a simple streaming program in Spark to receive text data streams on a particular port, perform basic text cleaning (like white space removal, stop words removal, lemmatization, etc.), and print the cleaned text on the screen. (Open Ended Question).

CODE, COMMAND WITH OUTPUT

```
# Install NLTK and download required data (run once)
!pip install nltk

import nltk
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')

from pyspark.sql import SparkSession
from pyspark.sql.functions import col, lower, regexp_replace, split, explode, udf
from pyspark.sql.types import ArrayType, StringType
from pyspark.ml.feature import StopWordsRemover
from nltk.stem import WordNetLemmatizer

# Initialize SparkSession
spark = SparkSession.builder.appName("TextProcessing").getOrCreate()

# Define your input lines
lines = [
    "Hello, I hate you.",
    "I hate that I love you.",
    "Don't want to, but I can't put",
    "nobody else above you."
]

# Create DataFrame from lines
df = spark.createDataFrame(lines, "string").toDF("value")

# Step 1: Lowercase and remove punctuation
df_clean = df.select(
    regexp_replace(lower(col("value")), "[^a-zA-Z\\s]", "").alias("cleaned")
)
```

```
# Step 2: Tokenize the cleaned text
df_tokens = df_clean.select(
    split(col("cleaned"), "\\s+").alias("tokens")
)

# Step 3: Remove stop words
remover = StopWordsRemover(inputCol="tokens", outputCol="filtered")
df_filtered = remover.transform(df_tokens)

# Step 4: Lemmatization using NLTK WordNetLemmatizer with UDF
lemmatizer = WordNetLemmatizer()

def lemmatize_words(words):
    return [lemmatizer.lemmatize(word) for word in words]

lemmatize_udf = udf(lemmatize_words, ArrayType(StringType()))

df_lemmatized = df_filtered.withColumn(
    "lemmatized",
    lemmatize_udf(col("filtered"))
)

# Step 5: Explode the lemmatized words and show results
df_lemmatized.select(
    explode(col("lemmatized")).alias("word")
).show(truncate=False)
```

```
(spark_env)thilak@kali: ~
Session  Actions  Edit  View  Help
  Downloading joblib-1.5.3-py3-none-any.whl.metadata (5.5 kB)
Collecting regex≥2021.8.3 (from nltk)
  Downloading regex-2026.4.4-cp313-cp313-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (40 kB)
Collecting tqdm (from nltk)
  Downloading tqdm-4.67.3-py3-none-any.whl.metadata (57 kB)
Downloading nltk-3.9.4-py3-none-any.whl (1.6 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.6/1.6 MB 4.4 MB/s  0:00:00
Downloading regex-2026.4.4-cp313-cp313-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (801 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 801.1/801.1 kB 4.2 MB/s  0:00:00
Downloading click-8.3.3-py3-none-any.whl (110 kB)
Downloading joblib-1.5.3-py3-none-any.whl (309 kB)
Downloading tqdm-4.67.3-py3-none-any.whl (78 kB)
Installing collected packages: tqdm, regex, joblib, click, nltk
Successfully installed click-8.3.3 joblib-1.5.3 nltk-3.9.4 regex-2026.4.4 tqdm-4.67.3

┌──(spark_env)─(thilak㉿kali)-[~]
└─$ python
Python 3.13.12 (main, Feb  4 2026, 15:06:39) [GCC 15.2.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> import nltk
... nltk.download('punkt')
... nltk.download('stopwords')
... nltk.download('wordnet')
... exit()
...
[nltk_data] Downloading package punkt to /home/thilak/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package stopwords to /home/thilak/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
[nltk_data] Downloading package wordnet to /home/thilak/nltk_data...

┌──(spark_env)─(thilak㉿kali)-[~]
└─$ PYSPARK_PYTHON=~/spark_env/bin/python pyspark
Python 3.13.12 (main, Feb  4 2026, 15:06:39) [GCC 15.2.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
26/05/02 15:16:19 WARN Utils: Your hostname, kali resolves to a loopback address: 127.0.1.1; using 10.0.2.15 instead (on interface eth0)
26/05/02 15:16:19 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/05/02 15:16:21 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
^[[B^[[B^[[B^[[B^[[B^[[B^[[B^[[B^[[B^[[B^[[B^[[B^[[B^[[B^[[B^[[B^[[B^[[B^[[B^[[B^[[B^[[B^[[B^[[B^[[B^[[BWelcome to

      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /__ / .__/\_,_/_/ /_/\_\   version 3.5.0
      /_/

Using Python version 3.13.12 (main, Feb  4 2026 15:06:39)
Spark context Web UI available at http://10.0.2.15:4040
Spark context available as 'sc' (master = local[*], app id = local-1777715182784).
SparkSession available as 'spark'.
>>>
```

```
(spark_env)thilak@kali: ~
Session  Actions  Edit  View  Help
>>> from pyspark.sql.functions import col, lower, regexp_replace, split, explode, udf
... from pyspark.sql.types import ArrayType, StringType
... from nltk.stem import WordNetLemmatizer
...
... lines = [
...     "Hello, I hate you.",
...     "I hate that I love you.",
...     "Don't want to, but I can't put",
...     "nobody else above you."
... ]
...
... df = spark.createDataFrame(lines, "string").toDF("value")
...
... df_clean = df.select(
...     regexp_replace(lower(col("value")), "[^a-zA-Z\\s]", "").alias("cleaned")
... )
...
... df_tokens = df_clean.select(
...     split(col("cleaned"), "\\s+").alias("tokens")
... )
...
... stopwords = {"i","you","that","to","but","the","a","an","is","are"}
...
... def remove_stopwords(words):
...     return [w for w in words if w not in stopwords]
...
... remove_udf = udf(remove_stopwords, ArrayType(StringType()))
... df_filtered = df_tokens.withColumn("filtered", remove_udf(col("tokens")))
...
... lemmatizer = WordNetLemmatizer()
...
... def lemmatize(words):
...     return [lemmatizer.lemmatize(word) for word in words]
...
... lemmatize_udf = udf(lemmatize, ArrayType(StringType()))
... df_lemmatized = df_filtered.withColumn("lemmatized", lemmatize_udf(col("filtered")))
...
>>> for row in df_lemmatized.select(explode(col("lemmatized")).alias("word")).collect():
...     print(row)
...
Row(word='hello')
Row(word='hate')
Row(word='hate')
Row(word='love')
Row(word='dont')
Row(word='want')
Row(word='cant')
Row(word='put')
Row(word='nobody')
Row(word='else')
Row(word='above')
>>>
```